Vector addition

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add displacement vectors and use a model to explain where an object ends up.

SUCCESS CRITERIA - I can...

1. Draw and label arrows tip to tail; keep their length and direction.
2. Draw the resultant from the original start to the final endpoint.
3. State net east-west and north-south motion with units and direction.

Core: teacher selects from 1-10. **Stretch:** 11-12 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

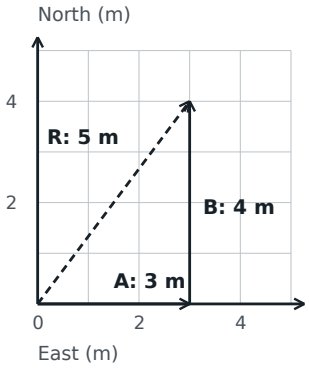
01 Bring back an earlier idea

A cart moves from 0 m to 8 m east in 4 s at a steady pace. Find its speed. Draw a velocity arrow showing its direction. Explain what the arrow tells you that speed alone does not.

Marzano check - circle: ① ② ③

02 Read and explain a worked model

A walker goes 3 m east, then 4 m north. A resultant is one arrow from the original start to the final endpoint.



Reference

$R_x = A_x + B_x$

$R_y = A_y + B_y$

x: east-west; y: north-south.

Worked reasoning

Path = 3 + 4 = 7 m.

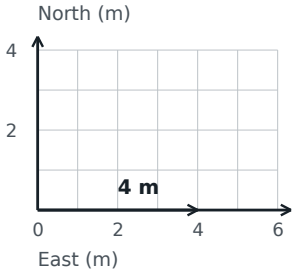
Displacement = given 5 m northeast.

Your job: Circle the start and endpoint. Explain why 7 m is distance, not displacement magnitude.

Marzano check - circle: ① ② ③

03 Finish the model

Walk 4 m east, then 3 m north. The first arrow is drawn. Add the second arrow from its tip and a dashed resultant from the original start.



Give the net eastward and northward motion. The straight-line length is 5 m: state displacement and direction. Also find distance traveled.

Marzano check - circle: ① ② ③

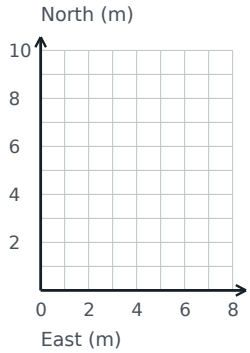
04 A trip that doubles back

Walk 6 m east, then 4 m west. Draw and label the route and resultant. Give distance and displacement. Explain which length follows the path.

Marzano check - circle: ① ② ③

05 Use a new scale

Walk 6 m east, then 8 m north. Draw both arrows and the resultant. Give distance and displacement. The straight-line length is supplied: 10 m.

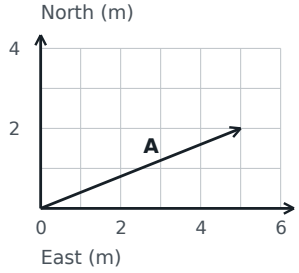


Show the start, turn and endpoint. Label each arrow with units. How does the drawing show that displacement magnitude is less than distance?

Marzano check - circle: ① ② ③

06 Add two diagonal arrows

A: 5 m east and 2 m north. B: 3 m west and 1 m north. A is drawn. Add B and R. State each net component and the general compass direction.



Marzano check - circle: ① ② ③

Use a ruler. Each grid square = 1 m. Read down the left column, then down the right.

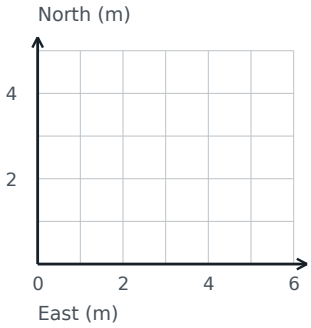
Vector addition

antocciphysics

Core 7-10 | Optional stretch 11-12 | Every grid square = 1 m.

07 Predict, draw, check

Walk 3 m east, 4 m north, then 3 m west. Predict what cancels. Draw the route and R. Give distance and displacement. Check your prediction against the model.

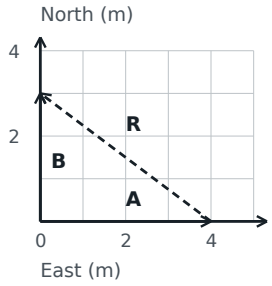


Marzano check - circle: 1 2 3

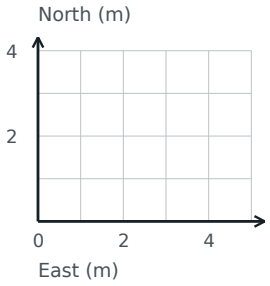
08 Find and repair the mistake

Sam adds 4 m east and 3 m north. His dashed R is wrong. Explain why and draw the correct route and resultant on the empty grid.

Sam's drawing: incorrect



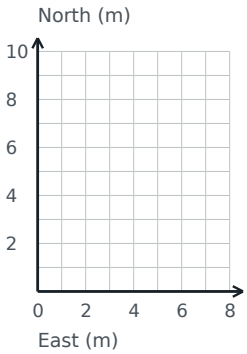
Your corrected model



Marzano check - circle: 1 2 3

09 One trip, three questions

A student walks 6 m east in 6 s, then 8 m north in 8 s. The direct start-to-finish length is 10 m. For each answer, decide whether to use path length or the start-to-finish arrow, then solve:
(a) Distance walked? (b) Displacement?
(c) Average speed for the whole trip?

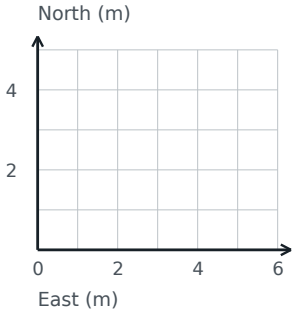


Draw and label the route and resultant. Show where total time enters your solution. Include units in all three answers.

Marzano check - circle: 1 2 3

10 Exit check - work independently

A goes 4 m east and 1 m north. B goes 1 m west and 3 m north. Draw tip to tail and add R. Give both net components and a general compass direction. Explain one check of your model.



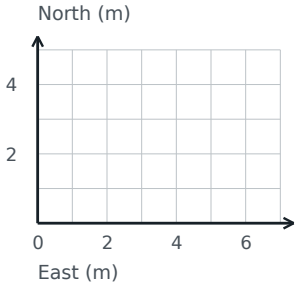
After feedback: Star a step that supports your rating. Explain any correction beside your work. Ask for help if the drawing and calculation disagree.

Marzano check - circle: 1 2 3

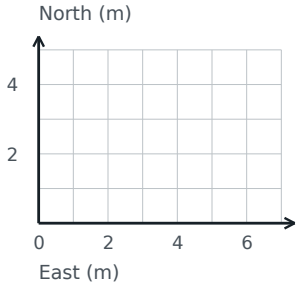
11 STRETCH: Design two routes

Finish 4 m east and 3 m north of your start. Design one route of length 7 m and another of length 11 m. Use only north, south, east or west legs. Label all lengths and draw each resultant.

Route 1: 7 m total



Route 2: 11 m total



Explain how distance can differ while displacement stays the same. Show a length check for each route.

Marzano check - circle: 1 2 3

12 STRETCH: Explore the limits

Point two 5 m displacement arrows in any directions. Draw the smallest and largest possible resultants, then one between those extremes. Estimate the intermediate resultant and explain why your limits make sense.

Keep one scale within each sketch. Label the original start, final endpoint and resultant. A ruler is allowed; no angle calculation is required.

Marzano check - circle: 1 2 3